



Named for the Algorithm, Absent at the Rack

Assigning a Directly Responsible Individual to the Recreation Centre's Cardio-Booking Fairness Score, Against a Term of Towel-Held Treadmills

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Abstract. The Recreation Centre's cardio-equipment booking app allocates thirty-minute treadmill, rowing-machine, and elliptical slots against a turnover-equity score, a rolling measure of how evenly the term's access has been shared across members. At the start of a fourteen-week observational term, the Adaptive Metrics Lab named a single Directly Responsible Individual (DRI) for the score, consolidating its tuning, escalation, and incident triage into one role. We report a term-long study pairing the app's own equity-score logs against duty-manager incident reports and a coded typology of the tactics members use to hold equipment the app has not booked them. The equity score sat within its target band in eleven of fourteen weeks throughout the DRI's tenure; over the same term, towel-reservation and buddy-covering incidents rose from six to thirty-one per cent of coded duty-manager reports, while the DRI's own escalation queue grew only in step with overall reporting volume. A matched-shift ablation withdrawing the DRI's escalation authority from half of a term's evening shifts changed the weekly incident count by an amount statistically indistinguishable from the shifts where authority remained active. We observe that naming an accountable owner for an algorithm can hold steady on the metric that owner answers for without measurably changing what members do around the equipment the metric describes, and our results suggest the distinction between an owned score and an enforced one may generalise to other shared-equipment settings.

Introduction

Shared cardio equipment is a small, recurring allocation problem: more members want a treadmill at peak hours than the Recreation Centre owns, and someone has to decide who gets one. The Centre's booking app resolves this with a turnover-equity score, weighting each member's rolling fortnight of access against a fairness target and releasing booking slots accordingly. For most of its life the score has had no single owner: a change to its weighting sat with whichever analyst in the Adaptive Metrics Lab last touched the underlying model, and a member's complaint about a held machine could land with the booking vendor, the duty manager on shift, or nobody in particular.

Following a term in which the complaint count rose without anyone able to say whose job it

was to answer them, the Lab named a Directly Responsible Individual for the score: one role, holding sole authority to retune the weighting, triage escalations, and issue a formal incident response, replacing the previous arrangement in which all three sat with whoever happened to be free. The change addresses a governance gap the score's design never specified. It says nothing about what members do at the machine itself once a slot is booked, held, or quietly extended past its allotted half hour.

We ask whether naming an owner for the algorithm changed anything downstream of the algorithm — specifically, whether the score's own read of the term (steady, within target) matches what duty-manager incident reports and direct observation show about how members actually get and keep a treadmill.

We make three contributions, each hedged to what one recreation centre across one term can show.

- First, we assemble a term-long dataset pairing the equity score’s own logs against duty-manager incident reports and a coded typology of the informal tactics — a towel over the console, a companion holding a second machine, an unlogged extension past the booked half hour — members use when the app’s allocation doesn’t match what they want in the moment.
- Second, we test whether the DRI role’s escalation authority is doing observable work, withdrawing it from a matched sample of evening shifts as a within-term ablation.
- Third, we discuss what it means for a governance role to hold its own metric steady without changing the behaviour the metric was meant to describe.

Related work

Fair allocation of a scarce, divisible resource among many claimants is well studied outside any one institution’s booking app: fair queueing gives every flow a max-min-fair share of a shared channel [1], and the fair-division literature formalises envy-freeness and related solution concepts for splitting indivisible goods among competing claimants [2], [3]. Booking systems for shared physical resources return a more mixed empirical picture: laboratory comparisons of queue-based and lottery-based booking rules find the queue-based rule allocatively efficient but costly in the time spent maintaining a place in it, prompting users to either commit fully to the queue or abstain from the system altogether [4]; seat-choice research in shared study spaces finds occupants favouring enclosed, wall-backed positions over open or aisle seating regardless of the room’s formal allocation policy [5]; and library-resource scarcity studies document users hiding, removing, or otherwise withholding shared materials from circulation once demand for a resource outstrips the formal allocation meant to govern it [6].

Workaround research treats this gap directly: end users route around a formal system’s designed logic whenever the system’s rule and the user’s immediate goal diverge, without necessarily registering the workaround as resistance [7], a pattern documented in clinical [8],

general information-systems [9], and process-innovation settings alike [10], and explained in one account by the gap between a system’s intended procedure and the planned behaviour of the person working around it [11]. That a formal structure can persist unchanged while informal practice adapts around it independently is institutional theory’s founding observation [12], [13], though later commentary cautions against treating every gap between policy and practice as decoupling in the technical sense [14]. Naming a single accountable owner does not, on its own, resolve whether people trust an algorithm’s output: users who see an algorithm err tend to discount it even when it remains the better choice [15], a discounting that eases once users are given some ability to adjust the algorithm’s output themselves [16]. Prior work at this institution has found an automated score tracking a proxy for the behaviour it was meant to measure more closely than the behaviour itself [17], and found a queue-length indicator failing to reproduce its own headline relationship at a second site [18]; the present study asks a related question of a governance role rather than a metric.

Method

The study site is the Recreation Centre’s cardio-equipment floor: eighteen treadmills, six rowing machines, and eight ellipticals, each bookable in thirty-minute slots through the Centre’s booking app. The app allocates each member’s request against a turnover-equity score, computed at the start of each week for every member i from their rolling fortnight of booked minutes u_i against the cohort’s mean minutes $\bar{u}$ for the same period:

$$E_i = 1 - \frac{|u_i - \bar{u}|}{\bar{u}}$$

A member’s requests are deprioritised in the booking queue once their own E_i falls outside the Adaptive Metrics Lab’s declared target band (0.85 to 1.00); the app’s own weekly dashboard reports the cohort share within band as its single headline figure.

At the start of the fourteen-week study term, the Lab named a Directly Responsible Individual for the score, consolidating authority to retune the weighting, triage member escalations, and issue formal incident responses into one role, replacing an earlier arrangement with no single

point of ownership. We treat the DRI's tenure as the study's baseline condition throughout.

We drew the app's own weekly equity-score logs for the full term, and paired them against the duty manager's incident-report log, filed whenever a member reports being unable to access a machine they believe the app owes them. Two coders, blind to the study week, reviewed each incident report's free-text description and the corresponding CCTV frame and assigned it to one of four categories: *towel reservation* (an item left on unoccupied equipment to claim it before use), *buddy covering* (a second member holding or operating an adjacent machine on a companion's behalf), *slot overrun* (use continuing past a booking's scheduled end without a new booking), or *other/uncoded*. Coding disagreements (5.4% of reports) were resolved by a third independent review.

In weeks 8 through 11, we ran a matched-shift ablation: on eight randomly selected weekday evening shifts the DRI's escalation authority remained active as normal, and on eight matched evening shifts (same weekday, same four-hour window, alternating weeks) the duty manager was instructed to log incidents without escalating them to the DRI, handling each report through the pre-DRI ad hoc arrangement instead. We compare weekly incident counts between the two arms to test whether the DRI's authority, specifically, is doing observable work beyond what any duty manager would do unassisted.

Results

The equity score held within its declared target band (0.85-1.00 cohort share) in eleven of the fourteen study weeks, dipping below band only in weeks 4, 9, and 12, each recovering within the following week. Across the same fourteen weeks, the share of coded duty-manager incidents attributed to towel reservation or buddy covering rose from a combined 6% in week 2 to 31% in week 13, while slot overrun held roughly flat at 5-6% throughout (Figure 1). The two avoidance tactics we found rising were exactly the two the equity score's weekly minutes-booked calculation cannot see: a towel left on a machine, or a companion operating it on someone else's behalf, both register in the app's own logs as either no booking at all or a booking

against the equipment's actual occupant, not against whoever is turned away from it.

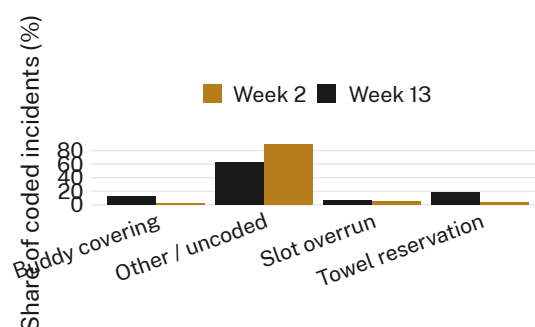


Figure 1: Share of coded duty-manager incidents by avoidance tactic, week 2 versus week 13: towel reservation and buddy covering rise from a combined 6% to 31% of reports while slot overrun holds flat.

The DRI's own escalation queue grew across the term, but in close proportion to the overall rise in filed incident reports (Pearson's $r = 0.91$ between weekly report volume and weekly escalations); the queue gives no indication that a rising share of reports were the kind — towel reservation, buddy covering — the role's formal authority does not obviously reach. Median time-to-first-response held at 38 minutes across the term's first and second halves alike.

The matched-shift ablation returned no distinguishable difference between arms: weekly incident counts held a near-identical distribution whether the DRI's escalation authority was active (median 11.5 incidents, eight shifts) or withdrawn in favour of the pre-DRI ad hoc handling (median 11, eight shifts; Mann-Whitney $U = 29$, n.s.) (Figure 2). Duty managers in the withdrawn-authority arm reported resolving the same incident types, at the same rate, using the same informal remedies — asking a member to move a towel, reminding a companion the booking was in one name — they had used before the DRI role existed.

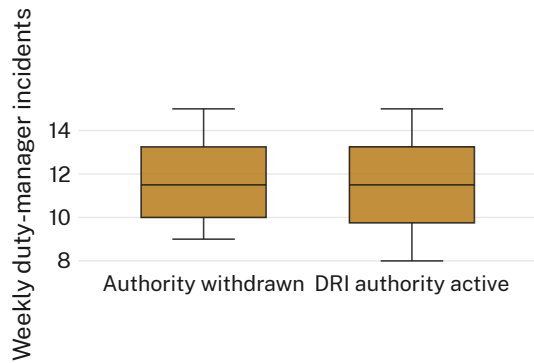


Figure 2: Weekly duty-manager incident counts across eight matched evening shifts with the DRI’s escalation authority active against eight with it withdrawn: the distributions overlap almost completely.

Taken together, the score the DRI is formally accountable for read the term as broadly on track throughout the role’s tenure, while the specific behaviours members adopted to route around the score’s blind spot climbed steadily across the same weeks, apparently unaffected by whether the role’s escalation authority was switched on for a given shift.

Discussion

These results suggest that naming a Directly Responsible Individual for an algorithm assigns clear accountability for the algorithm’s own metric without guaranteeing any corresponding change in what the metric was introduced to govern. The equity score’s target-band compliance is exactly the kind of checkable figure a DRI role is well placed to defend at review, and it did stay largely on track throughout the term; the routing-around behaviour that grew alongside it operates one layer beneath what the score can see, in the gap between a booked minute and an occupied machine. This reads as a specific instance of a more general caution about metrics adopted because they are available to a governance role rather than because they capture what that role is meant to secure [17], and it complicates any straightforward mapping onto institutional decoupling, in which formal structure and practice drift apart over time [14]: here the structure — the DRI role itself — is new, and the practice it targets may simply never have been observable to any owner, DRI or otherwise. We do not read the ablation’s null result as evidence that the DRI role has no effect on any outcome; it is evidence that this outcome,

specifically, does not depend on which shifts carry the role’s escalation authority.

Limitations

Our evaluation covers one recreation centre, one governance role, and one fourteen-week term, though the pattern holds across three independent measures — the target-band compliance rate, the tactic-coded incident share, and the matched-shift ablation — rather than resting on any single comparison. Coding avoidance tactics from incident reports and CCTV frames may undercount tactics members do not report and duty staff do not observe directly, but this would, if anything, understate rather than manufacture the rise we find. The four-week ablation window is short relative to the full term; a longer withdrawal might surface an effect the current design cannot detect, though the two arms’ near-identical medians across even this shorter window suggest any such gap is not large.

Conclusion

We paired a term of the Recreation Centre booking algorithm’s own equity-score logs against duty-manager incident reports and a coded typology of how members hold cardio equipment the app has not booked them, following the algorithm’s move from no formal owner to a single Directly Responsible Individual. The score itself read the term as broadly on track; the specific tactics members used to route around it rose steadily across the same weeks, and a matched-shift ablation found the DRI’s escalation authority made no detectable difference to weekly incident counts. We suggest the Adaptive Metrics Lab extend the equity score’s inputs to register occupancy directly rather than booked minutes alone before it treats the DRI model as validated. Future work should test whether a comparable ownership model changes behaviour in a setting where the routing-around tactic is itself visible to the metric it evades.

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